SHIVAM VINOD PRANJALE

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Senior Data Scientist | Industrial & GenAI Systems – John Deere India Ltd, Pune
Masters – IIT Madras

Senior Data Scientist with 8+ years of experience delivering industrial analytics, machine learning, and **production-grade Generative AI solutions** across manufacturing, automotive, and engineering domains. Currently leading **Agentic GenAI platforms** at John Deere to automate Theory of Operation authoring and technical documentation, achieving up to **50–60% productivity gains**. Strong expertise in **ML, DL, NLP, LLM-based systems, and industrial process analytics**, with a solid mechanical engineering foundation and an M.Tech from **IIT Madras**.

Core Skills

- **Industrial Analytics & ML:** Python, SQL, Pandas, NumPy, Scikit-Learn, Predictive Maintenance, CAE Analytics, Manufacturing Process Optimisation
- **Deep Learning & GenAI:** CNN, LSTM, NLP, LLMs, Generative AI, Prompt Engineering, Agentic AI Systems
- **LLM Systems & RAG:** LangChain, FAISS, OpenSearch, Retrieval-Augmented Generation
- **Tools & Platforms:** TensorFlow, Keras, Power BI, AWS, Jupyter, Git

Experience

Data Scientist I (Senior IC Role) - Analytics, Automation & PI Team - John Deere India Ltd. Pune (June 24 - Present)

- Architected GenAI-driven automation workflows in collaboration with technical authors and illustrators, streamlining industrial content authoring at scale.
- Designed and deployed an Agentic AI–powered Theory of Operation (ToO) automation platform for electrical, hydraulic, and mechanical systems, reducing authoring effort by up to 50%.
- Led SARC automation initiatives (including SIMPLE CI), enabling automated translation, quality checks, and resolution use cases, resulting in a ~60% reduction in turnaround time.
- Optimised enterprise knowledge bases and prompt strategies for hydraulic and mechanical domains, significantly improving grounding and first-draft accuracy of GenAI-generated technical content.
- Drove manual RCA automation and cross-team RCA forums, enabling data-backed corrective action recommendations and improved root-cause resolution efficiency.
- Authored and presented GenAI innovation concepts (Live Agentic Content Generation, Agentic ToO) at internal innovation and leadership forums, influencing enterprise GenAI adoption.

Data Scientist – Corporate Central Analytics Team - TATA Motors Ltd, Pune (April 23 – May 24)

- Delivered analytics and ML-driven solutions to improve operational efficiency and profitability across multiple automotive manufacturing use cases.
- Implemented end-to-end machine learning pipelines and Power BI dashboards to enable data-driven decision-making for enterprise stakeholders.
- Developed and deployed time-series forecasting and anomaly detection models on live sensor data, achieving a 70% reduction in preventive maintenance time.
- Mentored cross-functional teams on statistics, machine learning, and deep learning best practices, improving analytical maturity and solution quality.

Data Scientist – Smart Manufacturing & AI Team - Micron Semiconductors Ltd. Hyderabad (July 22 – Feb 23)

- Supported development teams in enhancing the Manufacturing Operating System (MOS) by contributing to visual and functional improvements for an Angular 14-based application.
- Analysed semiconductor manufacturing processes to identify process improvement opportunities using data science and statistical analysis techniques.
- Developed a proof-of-concept NLP solution leveraging user comment data from the MOS dashboard to surface actionable insights for process optimisation.

Machine Learning Consultant – General Motors - Engineering & Industrial Services, TCS, Bengaluru (July 20 – June 22)

- Delivered data analytics and machine learning solutions for CAE engineering workflows, accelerating simulation and design optimisation cycles.
- Developed prediction models for PM motor laminate stress, reducing optimisation iterations and cutting HPC usage by ~80%.
- Implemented NVH mode-shape prediction models for PM motor stators using material and manufacturing parameters, achieving a 60% reduction in optimisation cycles.

Technical Illustrator and Author – Indian Navy - Engineering and Industrial Services - TCS, Pune, MH (Mar2016 – June18)

- Contributed to the development of an engineering Learning Management System (LMS) for the Indian Navy by creating structured technical content for submarine systems.
- Developed high-quality technical illustrations using SolidWorks and Adobe Illustrator to enhance clarity and usability of engineering training material.
- Authored technical learning modules by analysing and extracting relevant information from complex submarine sub-system documentation provided by the client.

Academics

- Masters in Applied Mechanics - IIT Madras (2018-2020) CGPA – 8.81/10
- Bachelors in Mechanical Engineering - PRMITR Amravati(2011-2015) CGPA – 9/10

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Certifications

- GenAI
  - Prompt Engineering for ChatGPT
  - Gen AI – Gemini Pro
- Data / BI
  - Analysing and Visualizing Data with Microsoft Power Bi
  - Tableau 2020 A-Z: Hands-On Tableau Training for Data Science
  - Introduction to Data Science
  - MATLAB
- Statistics
  - Statistics for Data Science and Business Analysis
  - Deep Learning Pre-requisites
  - The Complete SQL Boot camp 2021

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